**10 October 2018 (Wednesday 9am)**

# Experiment 1

# Uniform Acceleration

Khyle Calpe (405016683)

Hirofumi Koichihara

**TA: Dhaivat Joshi**

# Worksheet

I. PLOT

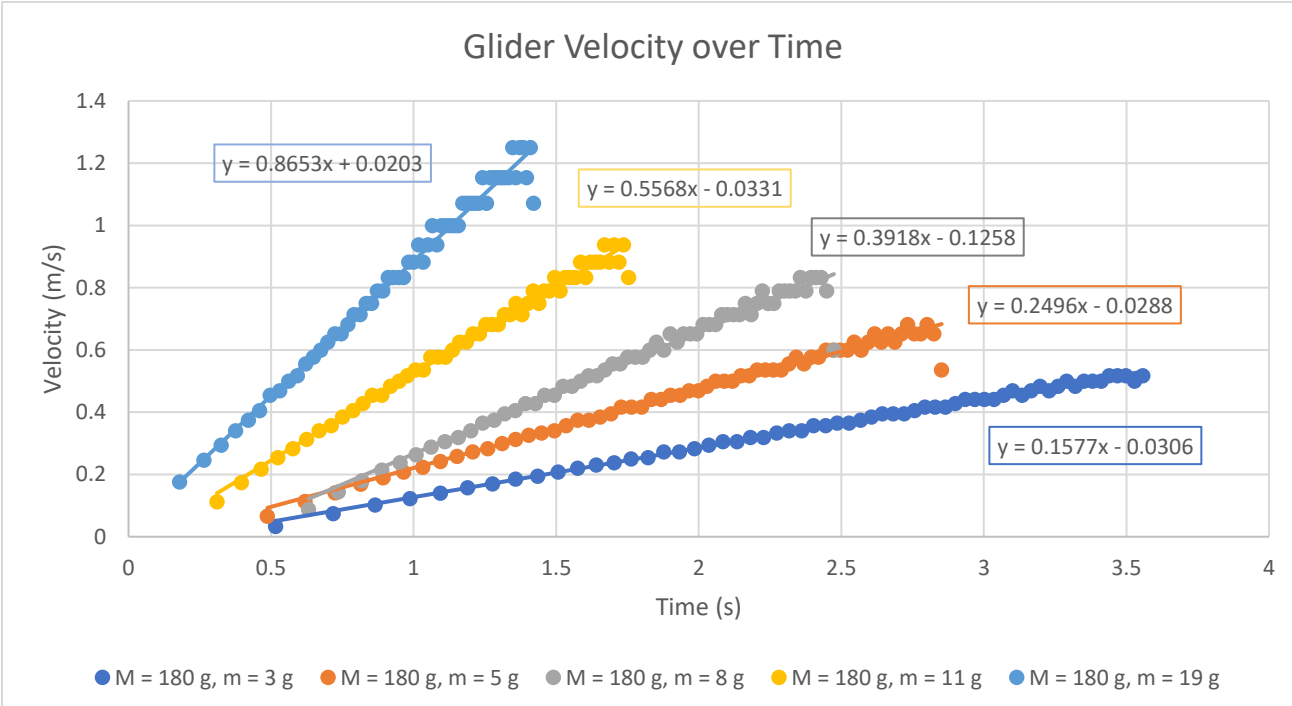


**Figure 1.1:** The data plot compares the velocity of a glider over time as the it moves on a near-zero friction surface of air while experiencing an applied force from varying masses (m = 3 g, 5 g, 8 g, 11 g, 19 g) falling from a pulley connecting the glider and the mass via a string.

The velocity of the glider is calculated using the following equation: $V_i = \frac{x_i - x_{i-1}}{t_i - t_{i-1}}$

The position $x_i$ of the glider at $t_i$ is calculated multiplying the distance the pulley's perimeter moves to transition from being unblocked to blocked $K_{SP} = (1.50 \pm 0.05)\ cm/block$ and the amount of times the photogate has been blocked and unblocked: $x_i = (K_{SP}) * (Block\ Count)$

Slopes of the runs based on weight mass:

$$m = 3\ g \quad \Rightarrow \quad 0.1577 \pm 0.0011\ m/s^2$$

$$m = 5\ g \quad \Rightarrow \quad 0.2496 \pm 0.0044\ m/s^2$$

$$m = 8\ g \quad \Rightarrow \quad 0.3918 \pm 0.0093\ m/s^2$$

$$m = 11\ g \quad \Rightarrow \quad 0.5568 \pm 0.0078\ m/s^2$$

$$m = 19\ g \quad \Rightarrow \quad 0.8950 \pm 0.0147\ m/s^2$$

## II. DATA TABLE

| Trial | Hanging Mass $m_{best}(g)$ | Glider Mass $M_{best(g)}$ | Fit Acceleration $a_{fit}(m/s^2)$ | Predicted Acceleration $a_{predict}(m/s^2)$ |
|---|---|---|---|---|
| 1 | 3.00 ± 0.05 | 180.00 ± 0.05 | 0.1577 ± 0.0011 | 0.1607 ± 0.0027 |
| 2 | 5.00 ± 0.05 | 180.00 ± 0.05 | 0.2496 ± 0.0044 | 0.2649 ± 0.0027 |
| 3 | 8.00 ± 0.05 | 180.00 ± 0.05 | 0.3918 ± 0.0093 | 0.4170 ± 0.0026 |
| 4 | 11.00 ± 0.05 | 180.00 ± 0.05 | 0.5568 ± 0.0078 | 0.5644 ± 0.0026 |
| 5 | 19.00 ± 0.05 | 180.00 ± 0.05 | 0.8950 ± 0.0147 | 0.9356 ± 0.0025 |

**Table 1.1:** Fit and predicted acceleration data of the glider while experiencing varying force due to different masses. Predicted acceleration was calculated using the equation: $a = \frac{gm}{m+M}$. Uncertainty $(\delta a)$ was calculated using the following uncertainty propagation formula:

$$\delta a = |a_{best}| \sqrt{\left(\frac{\delta m}{m_{best}}\right)^2 + \left(\frac{\delta g}{g_{best}}\right)^2 + \left(\frac{\sqrt{(\delta m)^2 + (\delta M)^2}}{(m+M)_{best}}\right)^2}$$

## III. DERIVATIONS

a. Acceleration of the Glider-pulley-weight system

Weight: $ma = mg - T$
Glider: $Ma = T$

$$ma = mg - Ma$$
$$a(m + M) = mg$$
$$a = \frac{mg}{m + M}$$

b. Uncertainty Propagation of the system

Uncertainty formula for sums and differences:

$$\delta f = \sqrt{(\delta x)^2 + \cdots + (\delta z)^2}$$

Uncertainty formula for products and ratios:

$$\delta f = |f_{best}| \sqrt{\left(\frac{\delta x}{x_{best}}\right)^2 + \ldots + \left(\frac{\delta z}{z_{best}}\right)^2}$$

The uncertainty propagation of the system can be derived by first addressing the uncertainty of the sum of $m$ and $M$. Then, the uncertainty can be substituted in the uncertainty propagation formula for products and ratios with $\delta a = 0.0003\ m/s^2$.

Part 1:

$$\delta(m + M) = \sqrt{(\delta m)^2 + (\delta M)^2}$$

Part 2:

$$\delta a = |a_{best}| \sqrt{\left(\frac{\delta m}{m_{best}}\right)^2 + \left(\frac{\delta g}{g_{best}}\right)^2 + \left(\frac{\delta(m + M)}{(m + M)_{best}}\right)^2}$$

$$\delta a = |a_{best}| \sqrt{\left(\frac{\delta m}{m_{best}}\right)^2 + \left(\frac{\delta g}{g_{best}}\right)^2 + \left(\frac{\sqrt{(\delta m)^2 + (\delta M)^2}}{(m + M)_{best}}\right)^2}$$

## IV. CONCLUSIONS

The plotted data of the velocity of the glider over time and the measured and predicted acceleration of the glider show that the acceleration increases as more force is applied on the system. The measured data from the fit-line of the data plot yielded data sets with precise margins of error. Likewise, the predicted data from the calculation of $a_{best}$ and the corresponding uncertainty propagation $\delta a$ yielded data sets with precise margins of error. For the first, second, and fourth trials, both the measured and predicted acceleration values fell within their respective margins of error. However, for the third and fifth trials, neither the measured and predicted values matched each other within their respective margins of error. Again however, the deviation of the values from the third and fifth trials did not amount to a significant degree to affect the precision of the data sets. Therefore, it is fair to say that the measured and predicted values of acceleration agree with each other.

The non-zero uncertainties for all five trials lead my group to believe that systematic errors within the experiment were present. For example, the uncertainties can be attributed to imprecise measuring tools, the non-zero friction surface below the glider, and the non-level surface of table. By improving the sensitivity and calibration of the measuring tools, the precision of the data sets will increase. By leveling the gradient of the surface and reducing the friction between the glider and the surface, the precision of the data sets will also increase. Overall, many improvements to the setup of the experiment can be implemented to reduce the uncertainties of the data sets to near-zero.

## V. EXTRA CREDIT

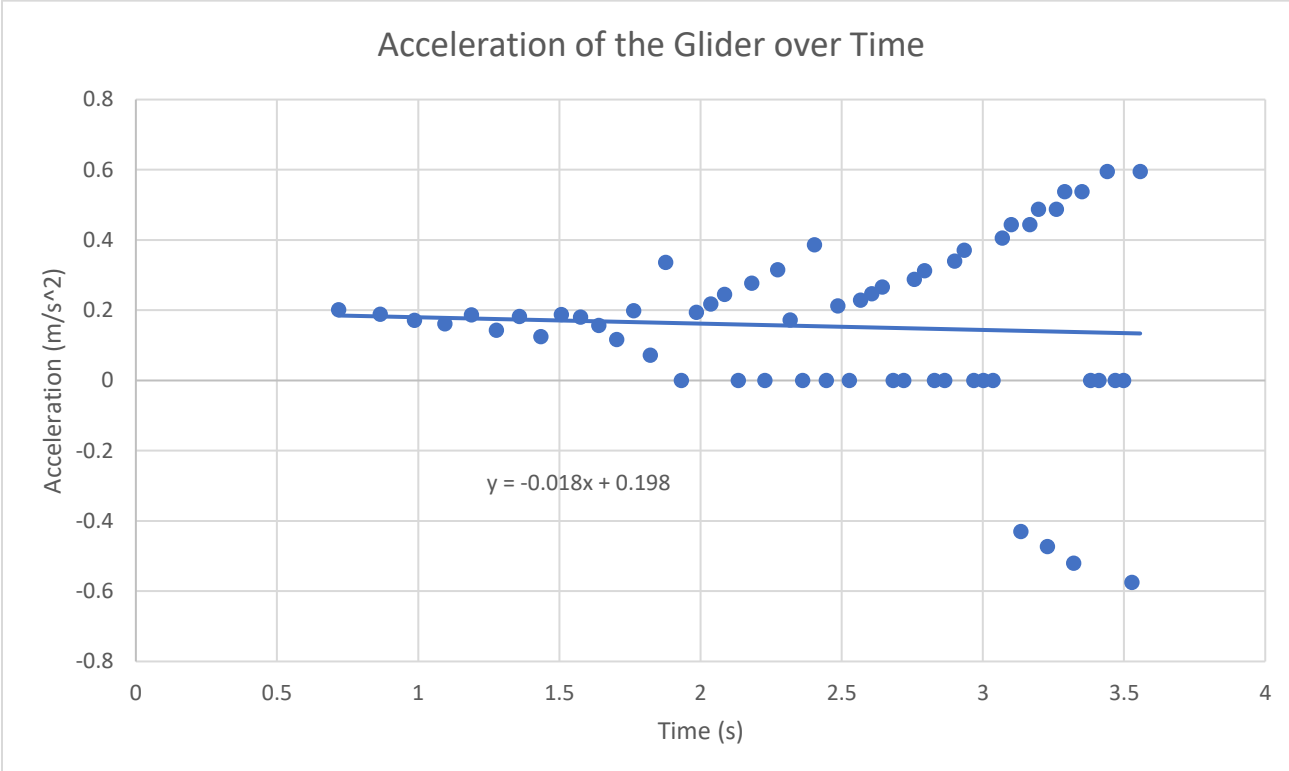


**Figure 1.2:** The data plot shows the differentiated acceleration over time of the glider. As the plot shows, as time progresses, the deviation of the acceleration from the trend line increases.

The deviation over time in the data is noise and is attributed to the non-constant value of acceleration. Within certain time intervals, the glider moves faster or slower. This results in positive and negative accelerations within the differentiated data set of the acceleration. This also means that, as the acceleration varies over time, the noise increases. This leads to the conclusion that the measurement analysis of the acceleration is a better method than a second order derivative measurement of the acceleration for both accuracy and precision of the data sets.

The acceleration of the system can be calculated by using the following equation:

$$a = \frac{1}{N}\sum_{i=1}^{N} a_i = 0.1536\ m/s^2$$

The uncertainty of the acceleration can be calculated by using the following equation:

$$\sigma_a = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(a_i - a)^2} = 0.2462\ m/s^2$$

# Mini-Report

## History of Recent Improvements in Photovoltaic Modules

Photovoltaic cells, or solar cells, are electrical devices that converts light energy into electrical energy via the collection of excited charge carriers. These photovoltaic cells are conventionally silicon-based due to the vast research on the element and its applications in addition to silicon's band gap for energy conversion and abundance in nature. Recent improvements in photovoltaic technology produced varying types of photovoltaic cells with ever-increasing efficiency in energy production. These improvements are best examined by discussing the advances in poly-crystalline silicon thin-film solar cells (pc-Si) in 2005 and the advances of perovskite solar cells in 2017.

Thin-film photovoltaics were developed to replace first-generation wafer silicon cells due to increases in energy-conversion efficiency. During 2005, peak efficiency for the most-efficient pc-Si reached 9.2% on a metal substrate. For practical purposes, glass sheets are conventionally coupled on silicon thin films due to glass's cheapness, transparency, stability, and abundance. Due to glass's drawback in its thermal stability, the most-developed pc-Si by CSG Solar coupled with glass reached an efficiency of 8.2%. To further increase photovoltaic conversion efficiency, the University of South New Wales developed three types of pc-Si thin-film solar cells by doping evaporated silicon via aluminum-induced texture (AIT): EVA, ALICE, and ALICIA. The three types producing varying results by applying the AIT process of deposition and crystallization at varying temperatures. These developments showed promising results in terms of energy conversion efficiency, cost efficiency, and scalability.

Further developments in photovoltaic technology expanded the materials used in photovoltaic module production such as perovskite. Perovskite is a class of materials sharing similar crystalline structures and also possess properties useful in photovoltaic energy production. By incorporating perovskite in solar cells, efficiency percentages have risen over 22.1%. With the increase in efficiency, the sustainability of the material must be considered.

Perovskite solar cells degrade over time due to the degradation of the material and the interface. The degradation of perovskite results from its hydrophilic nature which causes the material to easily absorb moisture from its environment and form hydrates. Furthermore, the perovskite can decompose into two salt products in a moist environment. Additionally, overheating can reduce the efficiency of the material and ion movement decreases the stability of the material. Lastly, the chemical reaction between perovskite and the electrode further decreases the stability of the photovoltaic cell. To alleviate this issue for practicality of the material for energy conversion, alterations to either the interface or the electrode have been proposed.

These recent advancements highlight the increased interest of photovoltaic technology research within the recent decade and the expected increase in photovoltaic cell efficiency. Currently, solar cells rely on a silicon base due to the impracticality of perovskite.

## Bibliography


1. A. G. Aberle *et al*., "Recent advances in polycrystalline silicon thin-film solar cells on glass at UNSW," *Conference Record of the Thirty-first IEEE Photovoltaic Specialists Conference*, 877-882 (2005).
2. X. Qin *et al*., "Recent progress in stability of perovskite solar cells*," *Journal of Semiconductors*, **38**, 011002 (2017).